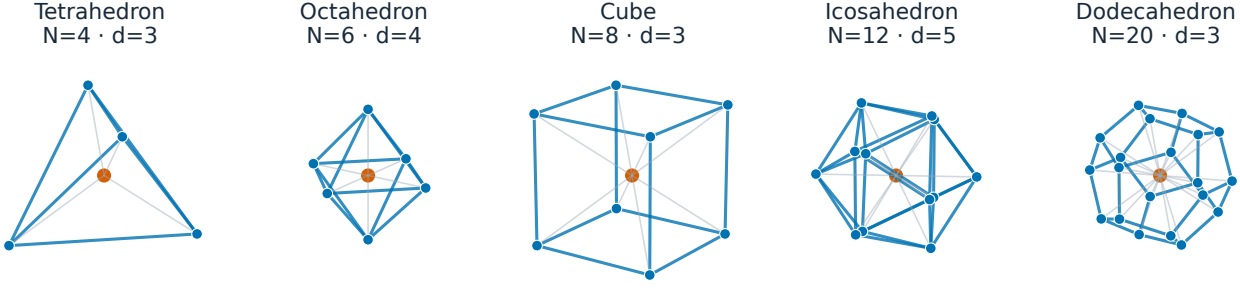
SYSTEM DIAGRAM

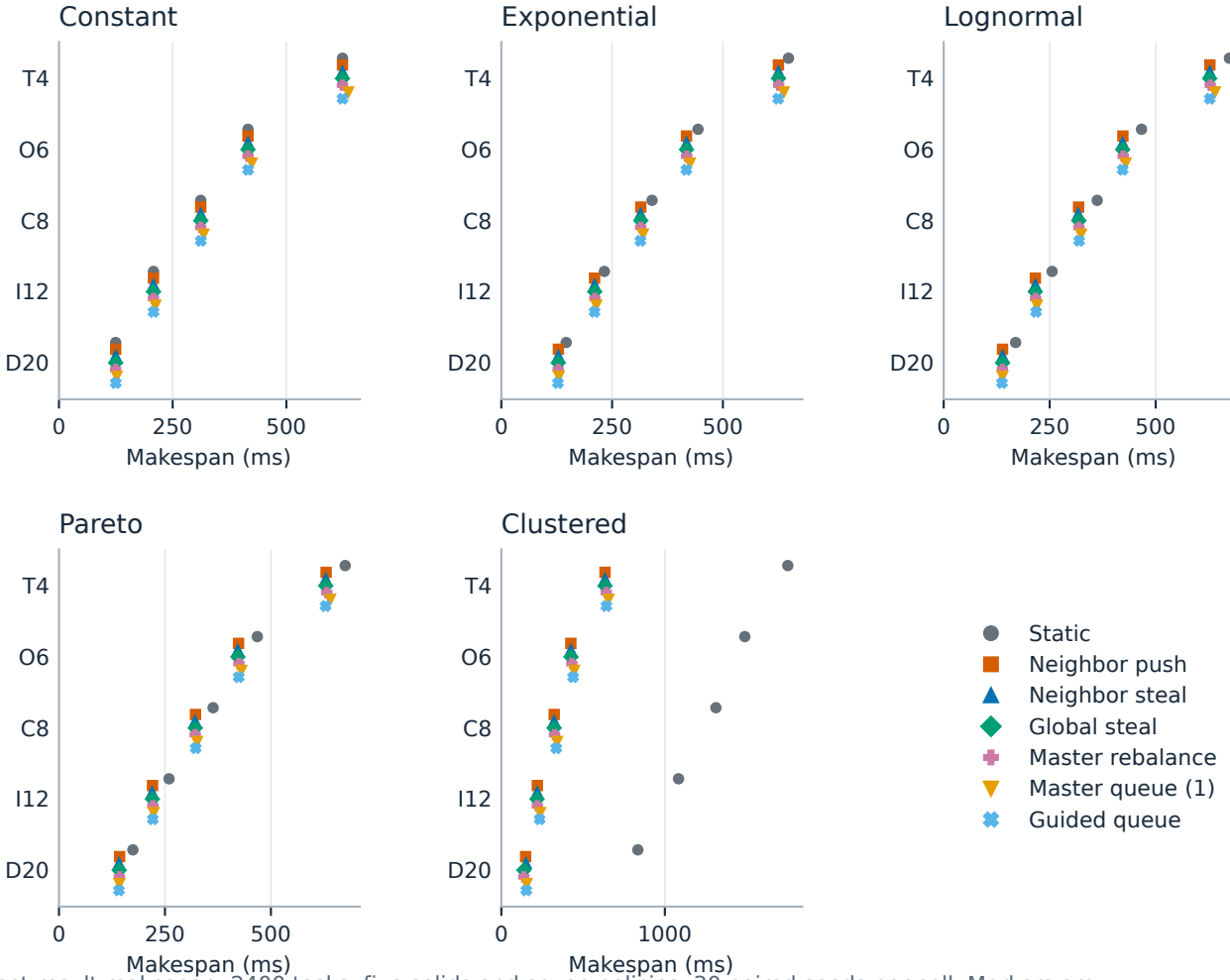
Five solids. One master. Only edge-adjacent peers.



Concept diagram: a red central master connects to every blue worker; dark peer links follow solid edges. N is the worker count and d is peer degree. Geometry specifies adjacency; all modeled links have equal bandwidth and latency.

EXPLORATORY SIMULATION

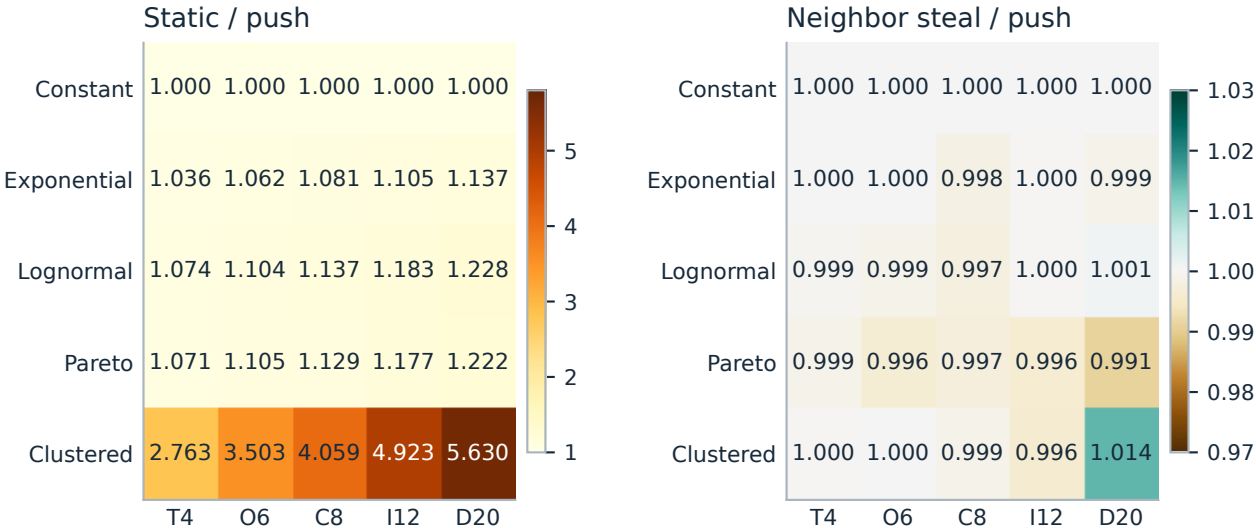
Redistribution helps imbalance; dynamic policies often overlap.



Last-result makespan, 2400 tasks, five solids and seven policies; 30 paired seeds per cell. Markers are means; horizontal bars are uncorrected 95% percentile-bootstrap intervals. Each worker count has its own physical network.

EXPLORATORY SIMULATION

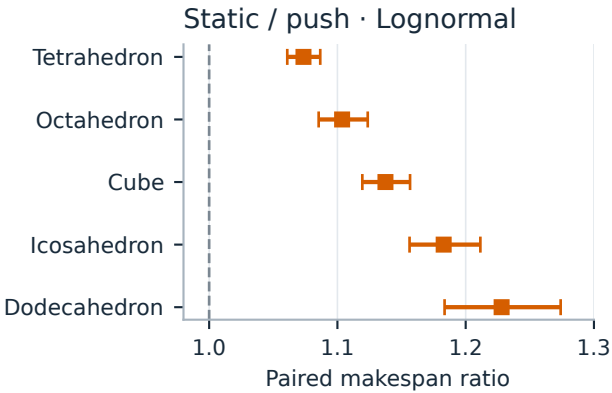
Large gains over static do not establish a push advantage.

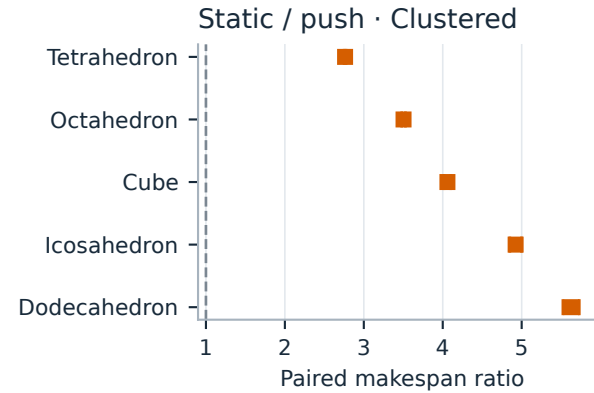


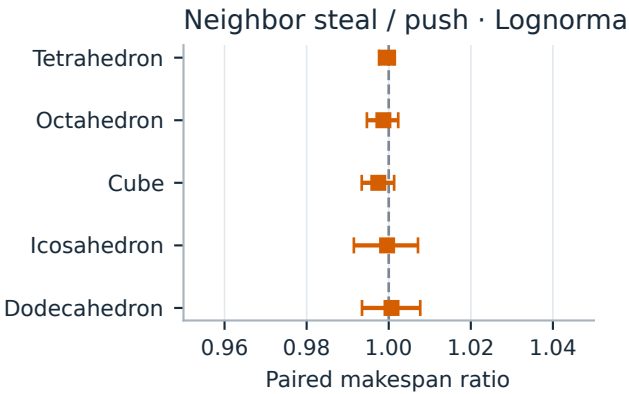
Mean paired makespan ratios from 30 seeds per cell; values above one favor neighbor push. Left: static/push. Right: neighbor-steal/push. Exact 95% intervals appear in paired_effects.csv and the companion interval plot. No equivalence claim is made.

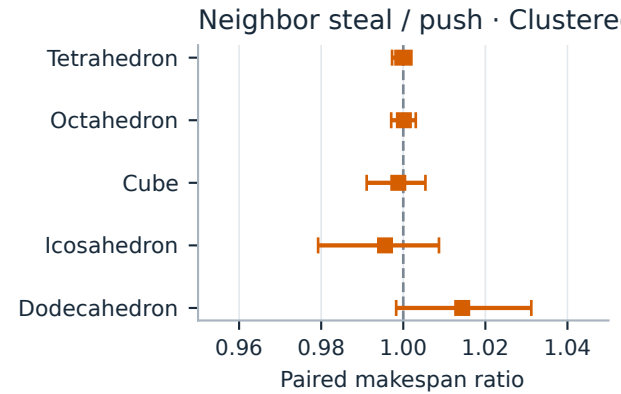
EXPLORATORY SIMULATION

Show the effect and its uncertainty, including null results.





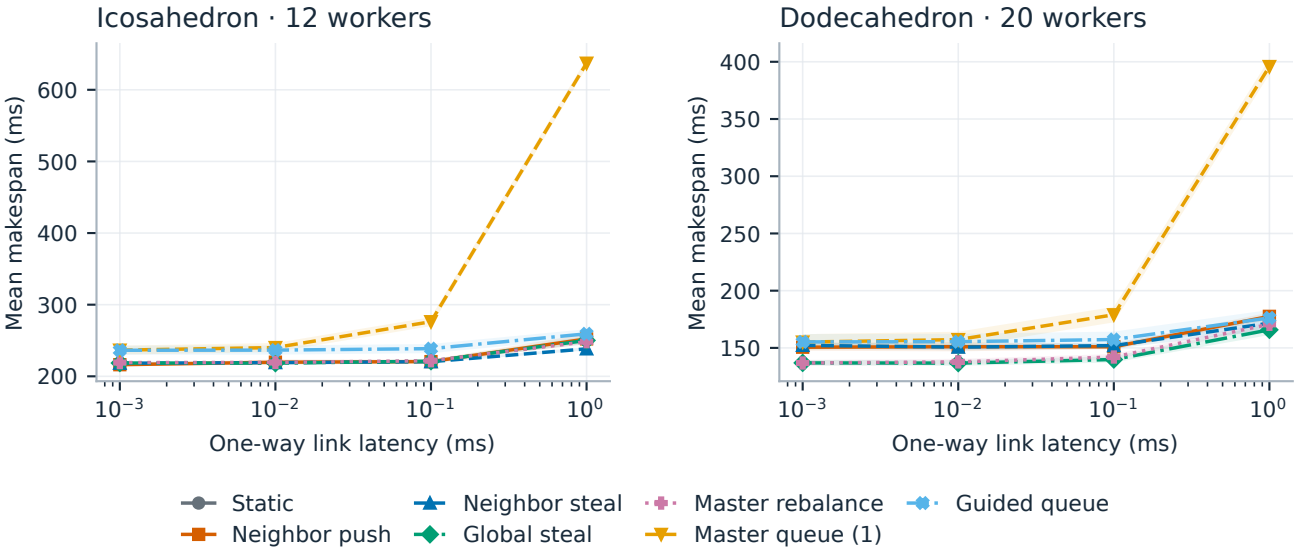




Mean paired ratios and uncorrected 95% bootstrap intervals, 30 seeds per solid. Dashed lines denote equal makespan. Lognormal is an independent-cost workload; clustered is a sorted-cost stress test. Interval overlap with one does not demonstrate equivalence.

EXPLORATORY SIMULATION

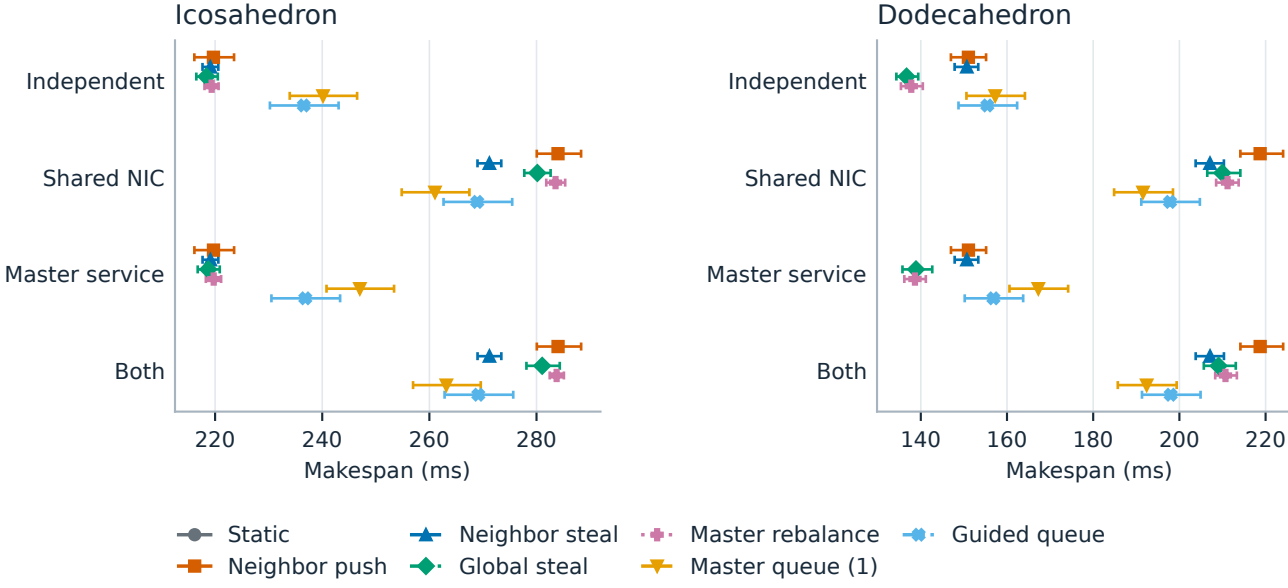
Fixed latency changes the scheduler tradeoff.



Clustered workload, 2400 tasks, 100 MiB/s; identical 15-seed subsets at every point. Lines show means; shaded bands show uncorrected 95% bootstrap intervals. Markers and line styles distinguish six dynamic policies. This is a latency sweep, not a full latency-by-bandwidth grid.

EXPLORATORY SIMULATION

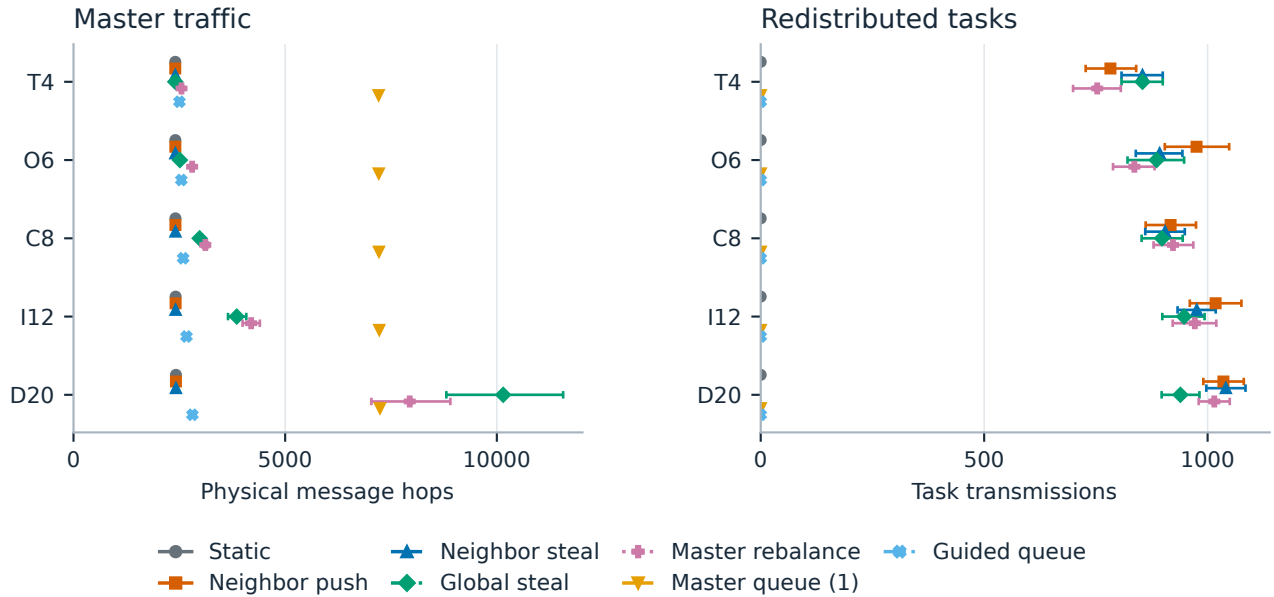
Independent links and a shared NIC are different regimes.



Clustered workload; the same 15 seeds in all four conditions. Points are means; bars are uncorrected 95% bootstrap intervals. Shared NICs serialize endpoint traffic. Master service adds 10 microseconds per incoming physical-hop message; 'Both' includes both constraints.

EXPLORATORY SIMULATION

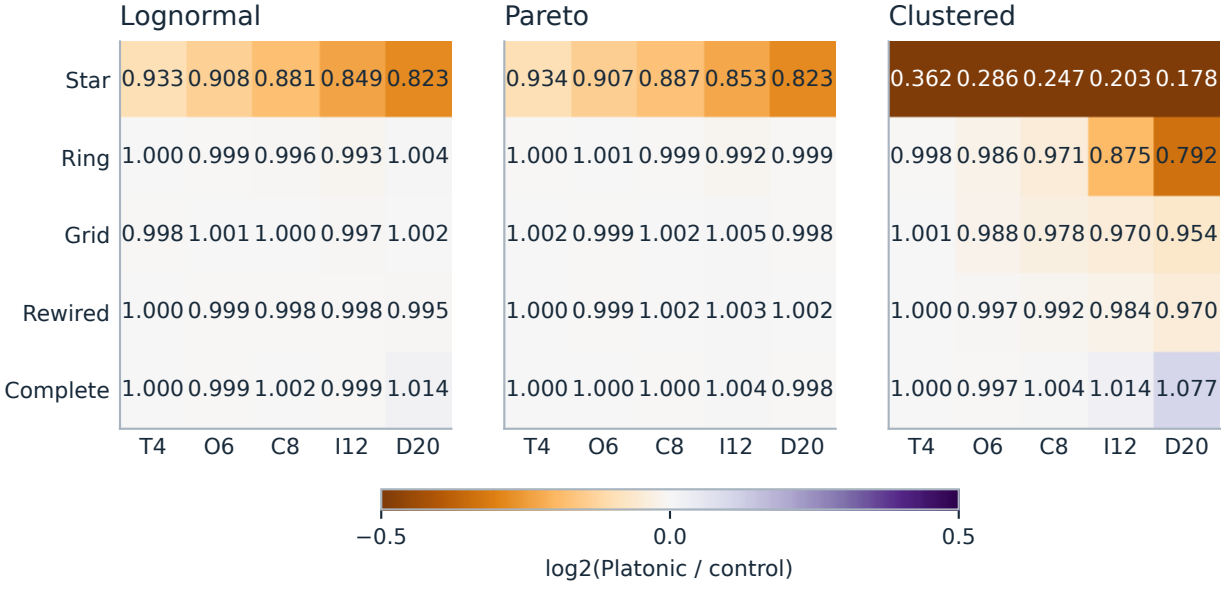
Local redistribution avoids extra master hops, not migration cost.



Clustered workload; 30 paired seeds. Means and uncorrected 95% bootstrap intervals. Master hops count initial transfers and all identified results. Migrations count transmissions, so the same ID can contribute repeatedly. Traffic includes control packets dispatched before the last-result stop.

EXPLORATORY SIMULATION

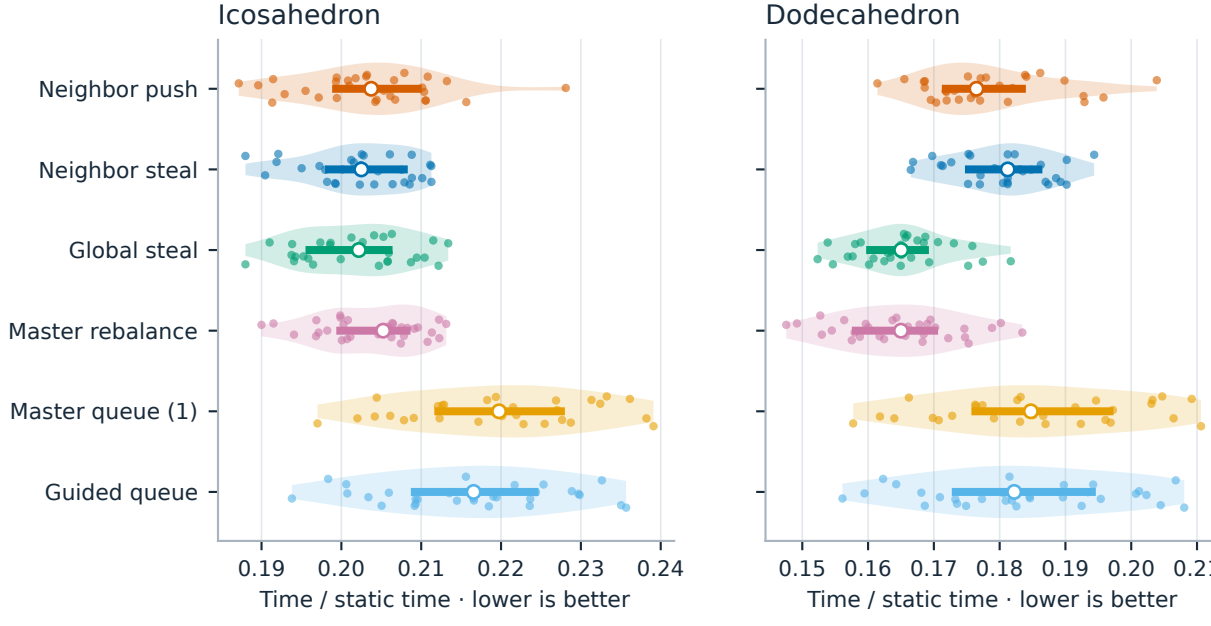
Compare peer topologies at the same worker count.



Neighbor push: mean paired Platonic/control makespan ratio, 30 joint graph/workload draws per cell. Below one favors Platonic connectivity. Only rewired controls preserve degree and peer-link count; the other rows have different link budgets. Color is clipped at +/-0.5 log2 units; labels give exact ratios.

EXPLORATORY SIMULATION

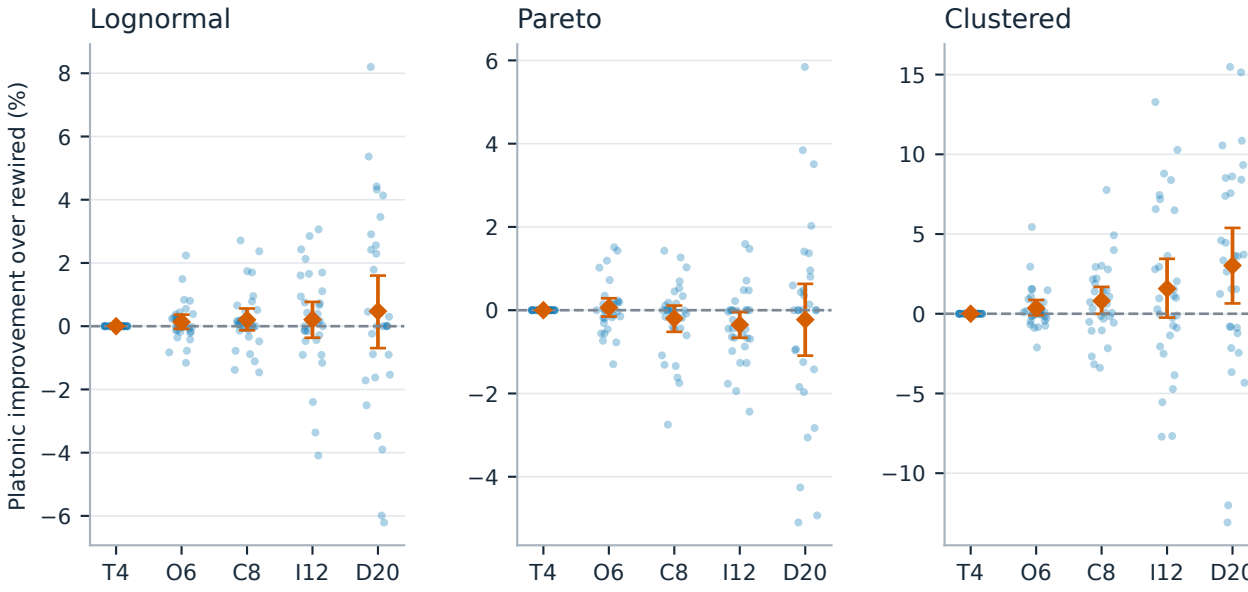
Keep every seed visible, not just the average.



Clustered workload; six dynamic schedulers, 30 paired seeds each. Every dot is one instance's time/static-time ratio. Violin shapes show the observed distribution; bold segments mark the interquartile range and white dots the median. Static is the reference, not a plotted competitor.

EXPLORATORY SIMULATION

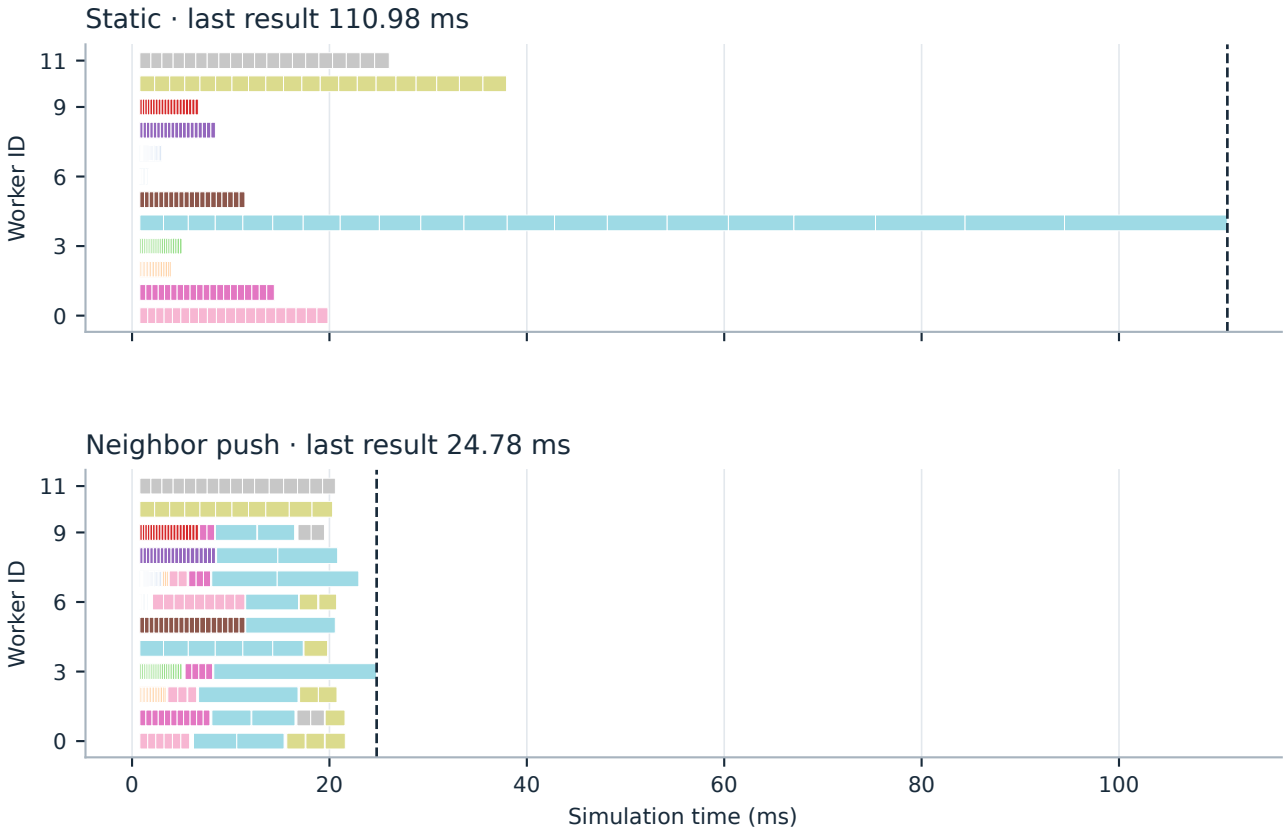
Degree-matched rewiring gives a more demanding control.



Neighbor push, 30 joint graph/workload draws per cell. Dots are paired percentage improvements, 100*(1-T_Platonic/T_rewired); positive values favor Platonic. Diamonds and bars show means and uncorrected 95% bootstrap intervals. Tetrahedral and octahedral rewiring is topologically equivalent up to labels.

EXPLORATORY SIMULATION

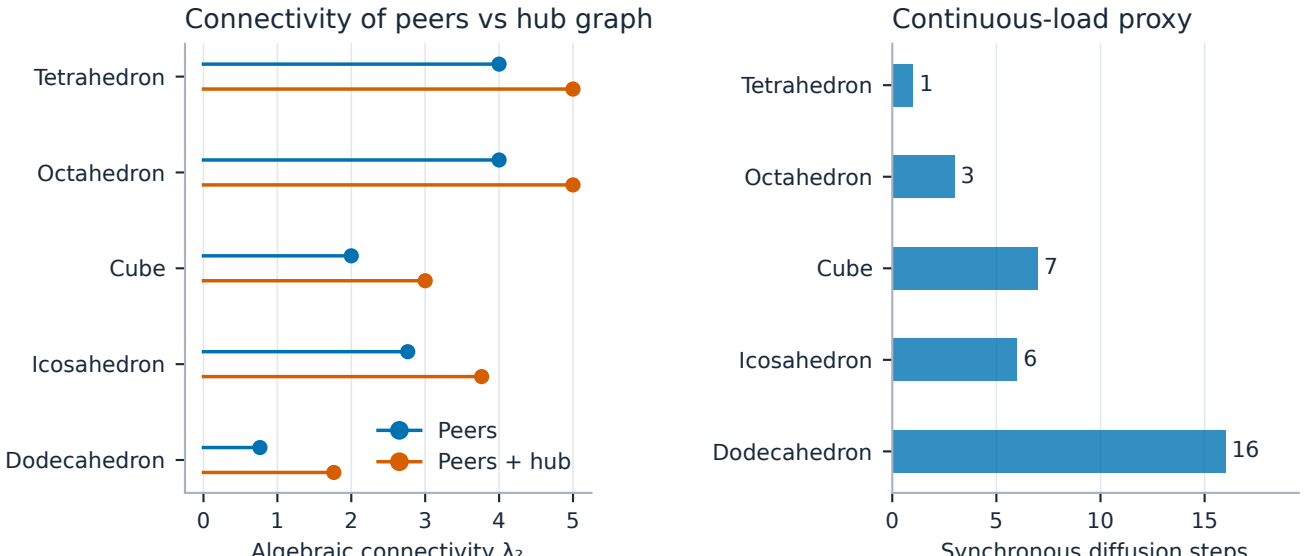
Task IDs make redistribution visible and auditable.



Illustrative icosahedral run: 240 clustered tasks, workload seed 2; this is not part of the primary benchmark. Colors identify original equal-count chunks, not policies. Each segment is one nonpreemptive task; the dashed line marks the last result collected at the master. Both runs retain every ID exactly once.

ANALYTIC GRAPH MODEL

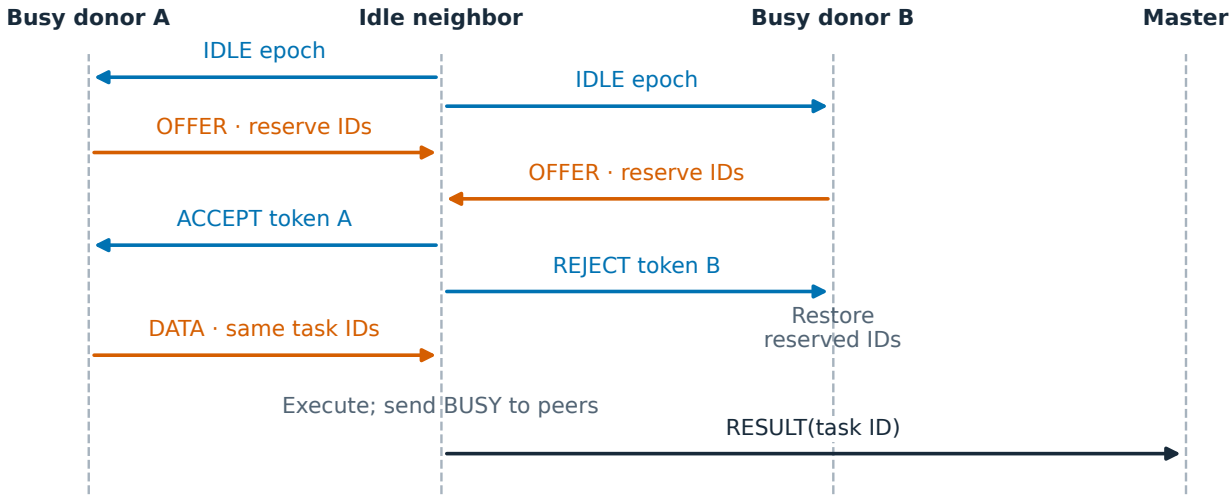
Graph structure supplies a testable model, not a speed guarantee.



Analytic graph calculations, not scheduler measurements. Left: algebraic connectivity lambda2 of the peer graph and graph plus master. Right: synchronous convex-diffusion steps sufficient for a 99% reduction of L2 deviation at the selected spectral parameter. Atomic idle-triggered scheduling is a different process.

IMPLEMENTED PROTOCOL · SCHEMATIC

Explicit messages resolve competing offers to an idle worker.



Schematic of the implemented neighbor-push protocol, not a timed trace. Donors reserve queued IDs before offering; the receiver accepts one offer and rejects a competing offer. Accepted inputs travel over a peer edge; identified results return directly to the master. No hub-assisted hint policy is implemented here.